

Meteor Madness

1 | Astroids, Meteors and Comets

- **Asteroids** : Small rocky bodies primarily in the asteroid belt between Mars and Jupiter; can sometimes cross Earth's path.
- **Meteors** : Bright streaks caused by small particles burning up in Earth's atmosphere.
- **Meteoroids** : The objects before they enter Earth's atmosphere.
- **Meteorites** : Meteoroids that reach Earth's surface.
- **Comet** : A small icy object in space that has a bright glowing tail when it gets close to the Sun. It is made mostly of ice, dust, and rocky materials, and orbits the Sun in a long, oval-shaped path.

2 | Missions & Space

1. **NASA's NEO Program** : NASA's Near Earth Object is a program that works on finding and tracking astroids and comets near earth using *telescopes* and missions to prevent possible impact.
2. **ESA's Space Guard** : the european space agency , is an organization that helps detecting and monitoring NEO that could threaten earth, they use ground base telescope like the *fleye telescope*, they collaborate with international partners (NASA) and they're developping missions to study NEO like *HERA*.
3. **OSIRIS-REX** : it's a NASA space craft launched to study the asteroid bennu, it collects samples and return them to earth (it collected samples in 2020 and is returning them in the coming years.)
4. **DART** : Double Asteroid Redict Test , Mission to tests *asteroid deflection techinques*, its main goal is to demonstrate that a spacecraft can intentionally impact an asteroid to change its trajectory. In 2022 DART collided with the asteroid moonlet Dimorphos which explained that the kinetic impact can alter the astroid's trajectory.

3 | Famous Astroids

1. **Chelyabinsk Meteor (2013):**
 - Diameter: ~20 m
 - Impact energy: ~0.5 Megatons of TNT
 - Impact height: Diameter: ~30 km

- Effects: hit Russia on February 15, 2013. It exploded in the atmosphere, causing damage and injuring over 1,500 people. It was a significant event highlighting asteroid risks.

2. **Tunguska Event (1908):**

- Approximate size: 50-60 m diameter
- Impact energy: 10-15 Megatons
- Effects: on June 30, 1908, in Siberia, Russia was a massive explosion caused by an asteroid or comet airburst, destroying thousands of square kilometers of forest with the energy of 10-15 megatons of TNT, and no crater was found.

3. **Meteor crater :**

- Diameter: ~50 m
- Age : 50 000 years
- Impact energy : ~ 10 Megatons
- Effects: this meteor created a crater about 1.2 kilometers and 170 m deep

4 | **Types of asteroids:**

1. C-type: Dark, carbon-rich asteroids that are very old and primitive, containing organic compounds and water.
2. S-type: Bright, rocky asteroids composed mainly of silicate minerals and metals, often found in the inner asteroid belt.
3. M-type: Metallic asteroids that are shiny and dense, primarily made of nickel-iron, and are thought to be remnants of shattered planetary cores.
4. NEAs (Near-Earth Asteroids): Asteroids whose orbits bring them close to Earth's orbit, posing potential impact threats and being important targets for study and exploration.

5 | **Types of meteors:**

1. Micrometeoroids: Tiny particles from space that create faint meteors or dust when they burn up in Earth's atmosphere.
2. Shooting Star (Meteor): Bright streaks of light caused by larger meteoroids burning up as they enter Earth's atmosphere.
3. Meteorite: Fragments of meteoroids that survive atmospheric entry and land on Earth's surface.
4. Impact Event: A collision where a large meteoroid or asteroid strikes Earth, potentially causing significant damage.
5. City Killer: A large meteoroid or small asteroid capable of causing destruction in urban areas upon impact.
6. Planet Killer: A massive asteroid or comet large enough to cause global or planetary-scale destruction if it hits Earth.

6 | Relation between both types

the type of asteroid can influence how it behaves when it enters Earth's atmosphere. C-type asteroids for example, are carbon-rich and darker in color therefore, when they enter Earth's atmosphere they can burn up and create a meteor. On the other hand, S-type asteroids are made of silicate materials and are more likely to survive the journey through the atmosphere and reach the ground as meteorites.

7 | Theoretical work (Asteroid dynamic)

7.1 | Initial equations

1. Kinetic energy

It helps determine how much damage or heat an object can cause during events like impacts or atmospheric entry.

$$KE = \frac{1}{2} m v^2$$

- m : Mass of the object
- v : Velocity of the object

2. Drag force

It slows the object down and increases with speed, surface area, and fluid density.

$$F_D = \frac{1}{2} \rho v^2 C_D A$$

- ρ : Fluid density
- v : Flow velocity relative to the object
- C_D : Drag coefficient
- A : Reference area

3. Heat flux

It is used to calculate the thermal load on an object during processes like atmospheric entry.

$$q = \frac{1}{2} \rho v^3 C_h A$$

- ρ : Fluid density
- C_h : Heat transfer coefficient (or heat transfer Stanton number)

4. Masse loss rate (due to ablation)

describes how quickly material is removed from a surface due to heating, often during atmospheric entry. It depends on heat flux, surface area, and heat of ablation.

$$\dot{m} = -\frac{q A}{H}$$

- $\dot{m}$: Mass loss rate due to ablation
- H : Heat of ablation (energy required to ablate a unit mass)

5. Deceleration

measure how quickly an object slows down, which is important for understanding effects like impact forces, braking, and atmospheric entry dynamics.

$$a = \frac{F_D}{m}$$

6. Remaining mass

$$m(t) = m_0 - \int_0^t \dot{m} dt$$

- m_0 : Initial mass
- t : Time

7. Thermal stress

for understanding how materials respond to temperature changes, especially in preventing structural failure or fragmentation in engineering and planetary entry scenarios.

$$\sigma = E \alpha \Delta T$$

- σ : Thermal stress
- E : Modulus of elasticity
- α : Coefficient of thermal expansion

- ΔT : Change in temperature

8. Stokes' law

calculates the drag force on small spherical objects moving slowly through a viscous fluid. It helps predict settling rates and resistance in fluids.

$$F_D = 6\pi\mu r v$$

- μ : Dynamic viscosity of the fluid
- r : Radius of the spherical object

9. Impact velocity

useful for predicting the energy released upon collision, assessing potential damage, and designing protective measures in fields like aerospace, defense, and planetary science.

$$v(t) = \sqrt{v_0^2 - 2 \int_0^t a_{\text{drag}} dt}$$

- $v(t)$: Velocity at time t (impact velocity)
- v_0 : Initial velocity
- a_{drag} : Deceleration due to drag
- t : Time

10. Impact energy

It helps assess the severity of an impact and potential damage caused.

$$E_{\text{impact}} = \frac{1}{2} m v_{\text{impact}}^2$$

- E_{impact} : Impact energy

7.2 | Updating the parameters

1. velocity :

$$v_{\text{next}} = v_{\text{current}} + a \Delta t$$

- v_{next} : Updated (next) velocity
- v_{current} : Current velocity
- a : Acceleration
- Δt : Time step

How to get a ?

$$a = \frac{F_D + F_g}{m}$$

where :

$$a = \frac{\frac{1}{2} \rho v^2 C_D A + m g}{m}$$

2. Area :

- **Circle:**

$$A = \pi r^2$$

- **Cone** (lateral surface area):

$$A = \pi r l$$

- **Cylinder** (total surface area):

$$A = 2\pi r h + 2\pi r^2$$

3. Drag coefficient :

- **Sphere:** $C_D \approx 0.47$

- **Cylinder:** $C_D \approx 1.0 \sim 1.2$
- **Cone:** $C_D \approx 0.5 \sim 1.0$

4. Atmospheric density :

- **10 km:** $\rho \approx 0.41 \text{ kg/m}^3$
- **20 km:** $\rho \approx 0.088 \text{ kg/m}^3$
- **30 km:** $\rho \approx 0.018 \text{ kg/m}^3$

5. heat transfer coefficient

- **C-type:** $C_h \approx 100 \sim 300$
- **S-type:** $C_h \approx 200 \sim 400$
- **M-type:** $C_h \approx 400 \sim 500$

6. Fragmentation approximation thermal stress

- If $\sigma \leq 0.1 \sigma_{\text{material}}$: ~3 fragments
- If $\sigma \geq 1.5 \sigma_{\text{material}}$: ~10 fragments
- If $\sigma \geq 3.0 \sigma_{\text{material}}$: 50+ fragments

7.3 | will we have an impact ?

impact threshold

- **C-type asteroid:** Impact if remaining mass $> 50,000 \text{ kg}$
- **S-type asteroid:** Impact if remaining mass $> 100,000 \text{ kg}$
- **M-type asteroid:** Impact if remaining mass $> 150,000 \text{ kg}$

8 | Trajectory calculation

kepler to cartesian

1. Compute the radius (orbital distance):

$$r = \frac{a(1 - e^2)}{1 + e \cos \nu}$$

2. Position in the perifocal (PQW) frame:

$$\vec{r}_{\text{PQW}} = \begin{bmatrix} r \cos \nu \\ r \sin \nu \\ 0 \end{bmatrix}$$

3. Velocity in the perifocal frame:

$$\vec{v}_{\text{PQW}} = \sqrt{\frac{\mu}{p}} \begin{bmatrix} -\sin \nu \\ e + \cos \nu \\ 0 \end{bmatrix}$$

where $p = a(1 - e^2)$

4. Rotation matrix from PQW to inertial frame (ECI):

$$\mathbf{R} = \begin{bmatrix} \cos \Omega \cos \omega - \sin \Omega \sin \omega \cos i & -\cos \Omega \sin \omega - \sin \Omega \cos \omega \cos i & \sin \Omega \sin i \\ \sin \Omega \cos \omega + \cos \Omega \sin \omega \cos i & -\sin \Omega \sin \omega + \cos \Omega \cos \omega \cos i & -\cos \Omega \sin i \\ \sin \omega \sin i & \cos \omega \sin i & \cos i \end{bmatrix}$$

5. Final position and velocity in inertial (Cartesian) coordinates:

$$\vec{r}_{\text{ECI}} = \mathbf{R} \cdot \vec{r}_{\text{PQW}}$$

$$\vec{v}_{\text{ECI}} = \mathbf{R} \cdot \vec{v}_{\text{PQW}}$$